

Abraham Kapambwe

C# Developer | Analytics Engineer | DevOps Engineer

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Professional Profile

Experienced full-stack developer with 14+ years of success in software engineering, big data analytics, and cloud-native DevOps. Adept in building scalable systems using .NET, designing real-time data pipelines with Apache Spark, and deploying infrastructure across Azure and AWS using Terraform and Ansible. Passionate about transforming data into intelligent systems through Microsoft Fabric, Power BI, and machine learning. Skilled in developing microservices with Docker and Kubernetes, creating knowledge graphs with NLP and graph databases, and applying geospatial analysis for real-world impact. Actively pursuing recognition as a Microsoft Most Valuable Professional (MVP) in Microsoft Fabric.

Technical Summary

Languages & Frameworks	C#, .NET (Core 3–8), Python, Blazor, ASP.NET MVC, JavaScript
Web & UI	React, Angular, ASP.NET Forms, Knockout.js
Data & Analytics	SQL, Couchbase, Apache Spark, Databricks, Power BI
DevOps & Infra	Azure DevOps, Docker, Kubernetes, Terraform, CI/CD Pipelines
Cloud & Platforms	Azure, AWS, Google Cloud, Windows Server, Linux
AI & ML	Graph DBs (Neo4j, Memgraph), NLP, ML Pipelines, TensorFlow, GANs

Work Experience

Software Developer | Derivco | Mar 2019 – Present

As part of a development team, I contributed to the design, integration, and deployment of scalable gaming and financial systems. My work focus on API development, payment integrations, DevOps automation, and big data processing on cloud platforms.

Key Contributions:

- Collaborated on integrating a wallet system with 20+ global payment providers including PayPal, Stripe, and cryptocurrencies.
- Supported the transition from legacy IIS applications to containerized microservices using Docker and Kubernetes.
- Participated in building Spark-based pipelines for processing logs via Kafka and implemented analytics in a Databricks Lakehouse.
- Contributed to automating multi-layered API testing and CI/CD pipelines in Azure DevOps.
- Applied Infrastructure as Code practices with Terraform across Azure.

Senior Developer | Ignition Group | Dec 2016 – Feb 2019

Contributed to the development and maintenance of key enterprise applications, including a customer contract management system, a billing engine, and a Twilio-based softphone platform for call centers. Led efforts to automate banking file integrations and ensured data consistency by feeding billing outputs into the warehouse database.

Key Contributions:

- Designed and implemented lifecycle workflows using Service Fabric for phone contracts.
- Automated multi-bank data processing (Standard, Capitec, Netbank, FNB) for financial operations.
- Built and maintained a robust billing system with continuous feature enhancements.
- Developed and deployed a softphone system using Twilio to improve call center efficiency.
- Collaborated closely with business stakeholders and agile teams to deliver system features.
- Applied SOLID principles to ensure clean, maintainable, and scalable software architecture.

Key Applications:

- Customer Contract Phone Management System
- Billing System
- Twilio Softphone Platform
- Magazine Subscription System

Software Developer | SA Web Design | Jun 2012 – Dec 2016

Contributed to the full development lifecycle of a comprehensive sports betting system, built from the ground up using agile methodologies. The system was implemented using the MVC

framework, with WCF as the service layer and a dynamic front end powered by ASP.NET and JavaScript.

Key Contributions:

- Developed interactive data capture interfaces based on wireframes and integrated them with WCF services.
- Designed relational database tables with constraints, stored procedures, and auditing triggers.
- Created operational and analytical reports using SQL Server Reporting Services (SSRS) and Report Viewer.
- Ensured codebase consistency using Team Foundation Server (TFS) and managed deployments to IIS and SQL Server environments.
- Resolved defects identified during testing to maintain application stability and reliability.

Mobile Developer / Web Developer | immedia | Sep 2011 – Jun 2012

Designed and developed mobile applications for tracking and managing trading shops, delivering both Android and Blackberry versions. Worked closely with the design team and client to align development with business requirements.

Key Contributions:

- Built Android app interfaces using Java and Eclipse, based on approved wireframes.
- Integrated mobile front ends with a Ruby-based backend API.
- Presented prototypes to clients, collected feedback, and implemented necessary adjustments.
- Managed version control and collaboration through GitHub.

Web Developer | Epages | Dec 2009 – Aug 2011

Delivered tailored web solutions for high-profile retail clients by developing and maintaining ASP.NET applications. Specialized in content management using Sitefinity and back-end integration with SQL Server.

Key Contributions:

- Built a cellular contract web application, aligning functionality with client requirements.
- Customized Sitefinity CMS to meet unique business needs and streamline content management.
- Maintained and enhanced client websites, including Game, Spar, and Dion Wired.
- Conducted client requirement reviews to ensure successful feature delivery.
- Managed the source code and version control to maintain development consistency.

Weekend Projects & Applied Innovation

I dedicate my weekends to exploring how emerging technologies can be strategically applied to help organizations solve complex problems, optimize decision-making, and create value through data-driven innovation. These self-initiated projects have allowed me to deepen my expertise across a range of industries, including banking, insurance, agriculture, healthcare, mining, telecom, and the public sector.

My domain knowledge spans key areas such as financial crime (e.g., trade-based money laundering, tax evasion), capital markets, accounting, auditing, and supply chain management. These initiatives have strengthened my ability to translate business challenges into scalable technology solutions.

- Microsoft Fabric: Lakehouse, Data Warehouse, Eventhouse, Data Pipelines, Real-Time Dashboards, Notebooks, Spark Jobs, Data Science Experiments
- Business Intelligence: Power BI (DAX, visualizations, semantic models)
- Data Governance: Data Catalog, Data Loss Prevention, Information Protection, Risk Management
- AI and Intelligent Agents: Azure AI Foundry, Microsoft Copilot Studio, Semantic Kernel, Phidata.
- Expert in designing and deploying AI agents using the Semantic Agent and Semantic Process Framework.
- Experienced in Multi-Agent Architectures and integrating Multiple AI Models for coordinated decision-making.
- Experienced in deploying business process workflows using the Semantic Process Framework with Dapr and .NET Aspire.

Detect Corruption Using Data Analysis Weekend Project

Developed a Microsoft Fabric data analysis pipeline to detect corruption in procurement using open contracting data. The GitHub repository shared contains all the artifacts used to process open contracting datasets, including Power BI reports and data AI agents created from the data. I am working towards creating a robust AI data agent using semantic agents and business process frameworks, which will transform natural language into SQL statements that can be executed on the data warehouse.

- Data Processing: Ingested, cleaned, and transformed open contracting data (JSON) into a Lakehouse and Warehouse environment.
- Risk Scoring: Calculated corruption risk scores for factors like single-bid contracts, repeated awards, short tendering periods, and unusual price variations.
- Power BI Reporting: Created interactive dashboards for visualizing corruption risk indicators.

- Technologies: Microsoft Fabric, Power BI, JSON, Data Lakehouse/Warehousing.
- GitHub: <https://github.com/Kapambwe/procurement-analysis>

Education & Certifications

- Microsoft Certified Professional (2011–2014)
- Coursera/Udemy: AI, Blockchain, TensorFlow, FinTech, CFA L1 Modules
- Udemy: Data Engineering (Azure, AWS, BigQuery), Power BI Projects
- NASA Earth Observation, Remote Sensing, Geospatial Intelligence

LinkedIn Articles & Publications

- [Using Microsoft Fabric to Combat Trade-Based Money Laundering](#)
- [Leveraging Microsoft Fabric to Combat Financial and Economic Crimes](#)
- [Revolutionizing Banking Operations with Microsoft Fabric](#)
- [Building an Analytic Platform on Microsoft Fabric](#)
- [Metadata-Driven Pipelines with Microsoft Fabric's Data Factory](#)